

Meal Type clustering in Daily Food & Nutrition Dataset using K-Means Clustering

1. Business Understanding

Objective:

The principal aim of the study is to use K-Means Clustering for meal types as classified by their nutritional values concerning the Daily Food & Nutrition Dataset. This classification should enable understanding dietary habits, optimize meal planning for individual users, and offer specific nutrition recommendations.

Key objectives include:

Meal classification: These should identify clusters of different types of meals such as breakfast, lunch, dinner, and snack by the nutritional profile.

2. Dataset Overview:

Dataset Used: *Daily Food & Nutrition Dataset*

<https://www.kaggle.com/datasets/adilshamim8/daily-food-and-nutrition-dataset>

Description: This dataset contains **10,000** records of **food intake data** from **1,000 unique users**, covering a wide variety of **food items** and their **nutritional values**.

3. Preparing Data

The following EDA steps are followed:

1. Load dataset.
2. Drop unwanted columns.
3. Handle missing values.
4. Encode categorical features.
5. Process datetime features (NA)
6. Remove outlier
7. Split into train & test sets.

4. Modelling:

Model Hyperparameters:

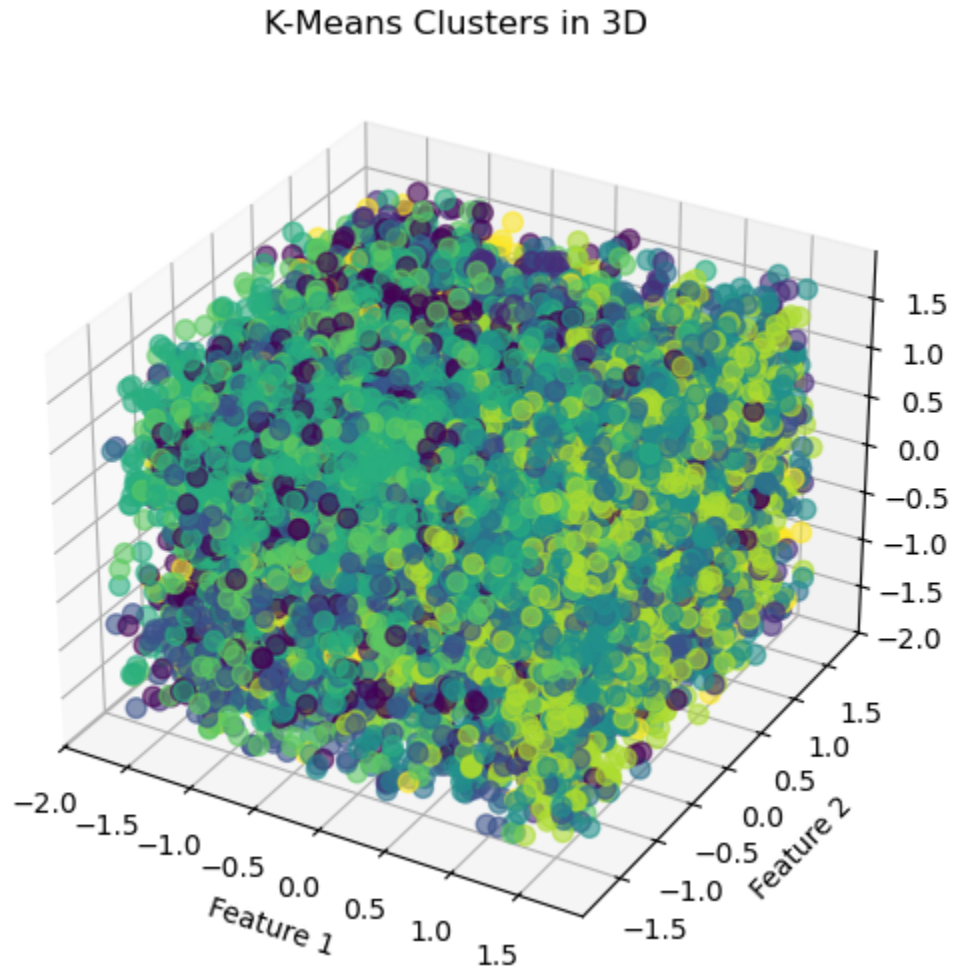
Number of clusters (k): has to explore for Maximum **Silhouette Score**.

The K-Means assigns a cluster label (**0-11**) to each **data point** based on the **closest centroid**.

5. Evaluation:

Evaluation Metrics:

Best **K** (optimal clusters) according to **Silhouette Score (0.08272176061074789): 2**



6. Observation:

- With hyperparameter tuning (**k value grid search**), we are able to get **Silhouette Score of 0.0839295427364303**
- After **dimensionality reduction**, we are able to get **Silhouette Score (after PCA): 0.04073845946537742**
- So, I am planning to explore more advanced algorithms like DBSCAN to get better results.